

Received 8 March 2025, accepted 25 March 2025, date of publication 8 April 2025, date of current version 18 April 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3558459

RESEARCH ARTICLE

Analyzing Relationships Between Success Factors of Project Management in Global Software Development

ANSAR SIDDIQUE¹, NOORULHASAN NAVEED QUADRI², NAIM AHMAD³,
AND HASHIM ELSHAFIE⁴, (Member, IEEE)

¹Department of Computer Sciences, Bahria University, Lahore Campus, Lahore, Punjab 54600, Pakistan

²Department of Computer Science, King Khalid University, Abha 62529, Saudi Arabia

³College of Computer Science, King Khalid University, Abha 62529, Saudi Arabia

⁴Department of Computer Engineering, College of Computer Science, King Khalid University, Abha 62529, Saudi Arabia

Corresponding author: Ansar Siddique (ansar.bulc@bahria.edu.pk)

This work was supported by the Deanship of Research and Graduate Studies, King Khalid University through the Large Research Project under Grant RGP2/179/45.

ABSTRACT In current era of technological advancement, global software development (GSD) is expanding to an increasing extent. This rise is driven by the advantages of low-cost development, access to expert resources, proximity to markets and clients, as well as options for innovation and the exchange of best practices. However, the various key aspects of GSD, including geographical dispersion, work ethics, time zone differences and cultural variability make project management more complex in a GSD setting. This study harnesses the critical success factors (CSFs) approach to identify the model for successful management of GSD projects. Sixteen CSFs have been determined using literature review, expert judgement, and exhaustive interviews. The interrelationship among the identified CSFs has been modeled using interpretive structural modeling (ISM) and cross-impact matrix multiplication applied to classification (MICMAC) analysis. The proposed model has been validated using detailed interviews, conducted with twelve professionals from industry and academia. The expert participants were carefully chosen based on criteria which include working experience, roles, and participation in GSD projects. This research computes CSFs of software project management in GSD by assessing their driving and relative strengths. Furthermore, these CSFs are categorized using MICMAC analysis. The previous research studies lack in exploring the mutual influence of CSFs and their hierarchical relationships. With the application of ISM and MICMAC analysis, the proposed study offers understanding of these interdependencies, providing great insights into their driving and dependent function. These insights enhance knowledge in GSD project management by allowing practitioners to prioritize key criteria and formulate potent management strategies.

INDEX TERMS GSD projects, CSFs, software project management, interrelationships, ISM, MICMAC.

I. INTRODUCTION

Global software development (GSD) “represents one of the emergent collaborative processes for developing a software product to be performed by team members belonging to different geographical locations” [1]. In recent times, GSD has grown as an accepted paradigm by plenty of software organizations. In fact, GSD had been able to attract such

widespread welcome due to the latest developments in international telecommunication at affordable costs combined with the ability to acquire appropriate skilled software people at relatively moderate costs. Access to a global talent pool remains one of the primary motivations for software companies to operate within GSD setting [2].

Benefits of GSD, such as 24/7 development cycles, cost efficiency, accelerated time-to-market, innovation through diverse expertise, shared experiences/best practices, are being pointed out more and more by the software industry [3].

The associate editor coordinating the review of this manuscript and approving it for publication was Olarik Surinta¹.

However, GSD is also associated with considerable technical and managerial challenges. Software development is a social task, whether collocated or distributed, and GSD magnifies the trust, communication, and coordination challenges over distances. Cultural, linguistic, and time zone distances complicate active knowledge sharing and team cohesion, thereby jeopardizing project success. Advanced knowledge, extended set of competencies, and special considerations are required for a project manager in a GSD context [3].

Effective project management is important for traditional software development and it becomes more significant in the context of GSD. Most organizations adopt GSD without preparing themselves to manage its challenges; thus, projects are doomed to fail due to inappropriate project management practices that do not fit the GSD requirements. The success of a global project is conditional on using project management strategies fitted to particular demands of the GSD environment.

The project management practices are usually different in developed and developing countries due to communication norms and work habits, organizational structures, and cultural frameworks [4]. The GSD projects are more vulnerable to threats such as language and cultural differences, the issues related to compliance, and geographical differences. These need to be identified and controlled with the help of some mechanism.

In GSD, effective project management ensures that the project is delivered timely, within the budget, and with assured quality. It is crucial to promote effective and constant communication amongst team members belonging to different geographical locations, time zones, and cultures so that the end product could be produced as per quality standard and client expectations. The critical success factors (CSFs) play a pivotal role in efficient management of GSD projects. These factors help the project manager in identifying the crucial areas that require the utmost attention and thus improve the probability of project's success [5]. Properly addressing CSFs greatly enhances project outcomes.

Although various CSFs (relating to project management) influencing the outcome of GSD projects have been identified in past research studies. However, these factors are mainly linked to general project management rather than to the challenges of project management in GSD [6]. The prior CSFs based models for managing GSD projects lack comprehensive coverage of critical aspects, such as addressing large geographical scope and integration with software project management frameworks. Most of the previous research studies have emphasized on identifying CSFs using literature review, surveys, case studies and various prioritization techniques such as pareto analysis and multi-criteria decision-making analysis techniques [4], [7], [8], [9], [10], [11]. These approaches provided a valuable understanding of key factors impacting the management of GSD projects. But these studies do not examine the interdependencies among identified factors. A structured approach to understanding these interrelationships is currently lacking in the literature.

Understanding interdependencies among CSFs is vital in GSD project management since these factors do not act in isolation, they impact and drive each other. Knowing these relationships assist project managers in prioritizing the key drivers, efficient allocation of resources, analyze root causes and manage major issues. Furthermore, diagnosing interdependencies facilitates collaboration among distributed teams and enables software organizations to adapt management strategies in accordance to their specific challenges.

This research, therefore, seeks to present a theoretically sound and practically relevant framework for the management of GSD projects that would address both the unique challenges of GSD and the deficiencies of the existing models. The study recognizes substantial CSFs and builds the relationship model among CSFs for a successful management of GSD projects.

The factors have structural associations with each other that may be pinpointed using contextual inquiry with the help of professional judgement.

The interpretive structural modeling (ISM) is best suited for the proposed research because it helps in developing a hierarchy among CSFs and defining significant driving factors along with their dependencies. ISM is a vast improvement upon the conventional prioritization methods as it interprets and models multifaceted interrelationships and improves the analysis of complex environments such as GSD, where communication, coordination, and risk management are deeply interrelated. Furthermore, the cross-impact matrix multiplication applied to classification (MICMAC), adds to ISM by classifying CSFs into different groups according to the driving and dependency power, thus eliminating confusion between base and outcome variables. Such classifications assist project managers focus on the most influential factors to achieve project success. Collectively these techniques go beyond simply identifying CSFs, they formulate a dependency based structured framework that is useful for practitioners.

Following objectives are lay out for this study.

(i) To determine the substantial CSFs for the successful management of GSD projects. (ii) To develop a model illustrating the hierarchy of CSFs as facilitators and outcome variables using ISM and MICMAC methodologies.

The rest of the paper is organized into the following sections. Section II depicts the research methodology. Section III represents the CSFs of software project management in GSD. Section IV presents the analysis and results. Section V presents the model validation done through in-depth interviews. Section VI is about the discussion and section VII presents the conclusion, limitations and future work.

II. RESEARCH METHODOLOGY

The first step of the research process is to identify all the CSFs presented in the literature relevant to software project management in GSD projects.

To make it certain that the most apposite and effective factors undertook, a systematic filtering process is applied

which is based on two key criteria: (i) frequency in literature and (ii) conceptual overlap. Factors that come out consistently in various studies were considered to ensure their significance in GSD project management. Simultaneously, overlapping factors are analyzed to remove redundancies without losing their core contributions. The reduction was not merely to manage complexity in ISM, but to ensure that all selected factors comprehensively represent the key aspects of GSD project management without duplication. This method supports the validity of proposed research by ensuring that the most important and widely recognized factors drive the ISM and MICMAC analysis.

The expert validation is a widely recognized approach in ISM based studies. So, in this study, expert driven approach is employed to conduct validation, certifying that insights were drawn from experienced professionals in GSD project management.

During brainstorming session, the decision makers (DMs) considered only those CSFs which have a significant role in the successful management of GSD projects. Such CSFs are described in the subsequent section of the literature review. The details of the DMs involved in brainstorming session are shown in Table 1. All the DMs have extensive industry experience, with five to ten years of experience in managing GSD projects. Curtailing the number of CSFs allows to handle the complexity in ISM because the large number of CSFs may cause huge matrices to be examined besides the difficulty in pairwise comparisons.

TABLE 1. Details of decision makers.

Sr#	Role of Participants	Position	Years of Experience in (Total/Management of GSD projects)
1	Decision Maker	Senior Manager	20/8
2	Reviewer	Senior Manager	19/8
3	Reviewer	Project Manager	15/6
4	Decision Maker	Project Manager	13/6
5	Decision Maker	Team Lead	11/6
6	Reviewer	Team Lead	10/5
7	Decision Maker	Team Lead	09/07
8	Reviewer	DevOps Manager	09/5
9	Decision Maker	QA Manager	12/7
10	Reviewer	Project Coordinator	07/05
11	Decision Maker	Academician	20/10
12	Reviewer	Academician	20/10

In the second stage, the ISM technique was applied to the short-listed CSFs identified in the first stage. The ISM technique employed a group decision-making approach to determine contextual relationships amongst variables, thus prone to individual biases. In order to reduce the subjective effects, some measures were adopted which are as: The GSD domain experts ranging from senior project managers to software engineers and academicians were involved to ensure the balanced representation of perspectives. Secondly, to reach

a final decision, multiple rounds of expert feedback were taken in order to eliminate personal bias. Finally, relationships between factors were established with the majority vote instead of a singular view, which lowered the effect of biases.

The integration of interactive learning processes and the structural assembly of relevant variables are features of ISM [12]. It utilizes conceptual, theoretical, and computational leverage to demonstrate the contextual links between variables by drawing on the foundations of interactive learning and graph theory. Thus, ISM creates connections between facts to build a tangible model. It has been applied across various fields, including engineering, computer science, decision science, social science, cloud migration, mobile learning, and e-learning [12], [13], [14], [15].

The effect of one variable on another can be measured via ISM modeling. A set of DMs, can assess the interaction between two variables. A complex structure frequently has to be made simpler since the link is not clear-cut. The output of the ISM model is presented as a diagram depicting the relationships between different variables. Based on the relationships, a collection of matrix tables called the structural self-interaction matrix (SSIM) is produced. A pairwise comparison of two variables serves as the foundation for the SSIM formulation. The potential associations that DMs looked for in pairs of variables, like x and y , are indicated by the symbols V, A, X, and O, respectively. The reachability matrix is then created using this SSIM. These relationships can be x influences y , y influences x , x and y impact mutually, or x and y do not influence each other.

In a nutshell, the ISM modeling process consists of several steps, including identifying the variables of interest, constructing the conical matrix, dividing the reachability matrix, adding transitivity to the SSIM, and making the digraph. Once the matrices are created, the driving power and dependence values are computed. Such driving force and dependence values are utilized to categorize variables using MICMAC Analysis. To determine the set of variables, multiple interview sessions are conducted. In due course, a virtual online session is arranged to develop the relationships between the variables. The ISM and MICMAC analysis is performed using the SmartISM software, a widely accepted tool for the implementation of ISM and MICMAC methodologies [12], [16]. This software assists in automating complex mathematical processes, enabling a focused analysis on the identification of key relationships among variables. The analysis and findings are shown in the subsequent section.

III. CSFs OF GSD PROJECT MANAGEMENT THROUGH LITERATURE REVIEW

The adoption of GSD has increased recently due to its several benefits such as access to skillful resources at affordable cost, flexibility and innovation. The successful execution of GSD projects necessitates effective management approach. This research employs the CSFs approach to effectively manage the GSD projects. CSFs are vital actions, organizations performed to achieve competitive advantage. Rockart et al. [17]

has defined CSFs as “the limited number of areas in which results, if they are satisfactory, will ensure satisfactory competitive performance for the organization”. CSFs may help each type of organization in achieving desired outcomes. The concept of CSFs is gaining importance for the successful management of GSD projects. Several prior research studies have identified CSFs for the management of GSD projects. However, these studies focused on the limited set of CSFs rather than presenting full insight for project managers working in GSD setting. To identify CSFs in the area of software project management of GSD, an extensive literature review was conducted in our previous study [11]. Consequently, 45 research papers in total, published during 2010-2022, were selected to extract CSFs. This resulted in a total of 36 CSFs, which were reduced to 16 CSFs by eliminating those CSFs with limited support in the literature.

This study advances the prior research by specifically identifying interdependencies among CSFs applying ISM and MICMAC analysis, a perspective that is not thoroughly investigated in existing GSD project management literature. Through these methods (ISM and MICMAC), the interrelationships among CSFs were examined to evaluate, prioritize and classify them systematically. In a group decision making process, with the help of a brainstorming session and feedback solicited from DMs, the 16 most substantial CSFs were selected to manage the complexity in ISM methodology. During this session, CSFs were also ranked according to their mutual influence and dependence. The CSF variables, together with referenced articles, ranging from the most independent to the ultimate desired CSF fitting system for the successful management of GSD projects, as shown in Table 2. In the following paragraphs, these CSFs are elucidated.

TABLE 2. CSFs obtained through literature.

Sr#	CSFs	References
1	Productive communication skills	[1][3][4][6][7][8][51]
2	Adeptness in negotiation	[25][26][27][28]
3	Frequent communication	[31][32][33][34][51]
4	Effective coordination and control	[1][2][3][4][5][6][8]
5	Active involvement of users	[2][6][23][36]
6	Collaboration among team members	[1][3][4][8][20][21][22][51]
7	Proficient software developers	[2][4][8][40]
8	Appropriate formation of team	[41][42][43][44]
9	Roles and responsibilities of team members	[1][3][4][7]
10	State of the art technological infrastructure	[2][4][6][8][38][39]
11	Fostering cultural awareness	[1][5][6][7][51]
12	Sharing project knowledge	[45][46][47][48]
13	Managing time zone differences	[3][23][27][28][49]
14	Core project management processes	[3][23][27][28][30]
15	Tracking project progress	[4][5][8][23]
16	Language differences	[51][52][53][54]

A. PRODUCTIVE COMMUNICATION SKILLS

Productive communication skills are very important in GSD, where software development teams varied in respect of

geographical, cultural, and linguistic perspectives. Clear and concise communication confirms that software requirements and feedback are correctly conveyed, diminishing ambiguities and misunderstandings. Adaptability to various communication styles develops good working rapport and trust among distributed team members. Besides, cultural sensitivity and attention to inclusiveness build up a collaborative environment that results in better productivity and fulfilling the project's objectives more efficiently [48], [49], [51].

B. ADEPTNESS IN NEGOTIATION

GSD negotiation adeptness refers to the ability to negotiate cultural, linguistic, and organizational differences in such a way that the interests of all relevant stakeholders are aligned. It means clear communication, active listening, and the ability to understand different priorities across geographically dispersed teams and clients. Effective negotiators will find ways to address challenges such as conflicting time zones, differing work practices, and cultural norms in a way that benefits the agreement of both parties. All these have to be maintained through critical dispute resolution skills, real expectation setting of the project, and effective collaboration that are critical aspects for maintaining productivity and achieving objectives in a globalized software development environment [10], [32].

C. FREQUENT COMMUNICATION

Communication frequently arises in GSD to bridge gaps in geographical, cultural, and time zone differences between distributed teams. It makes sure that the information exchange flows uninterrupted, allowing the team members to clear up the requirements, align project goals, and resolve issues then and there. In this way, misunderstandings, which might turn up because of geographical separation, will be reduced. It allows for the use of different communication tools, such as video conferencing, instant messaging, and collaboration platforms for real-time interactions that help in building trust and collaboration. Regular updates and structured communication practices ensure the visibility of projects down to the granular level and make sure all stakeholders are kept informed and aligned despite the distributed environment [34], [51].

D. EFFECTIVE COORDINATION AND CONTROL

Effective coordination in GSD requires managing distributed teams of people across geographical, cultural, and temporal boundaries to meet project objectives for efficiency. Coordination ensures that team members align their activities, share knowledge, and work collaboratively despite physical separation—often through the facilitation of communication tools and shared repositories. Mechanisms of control will monitor progress toward and the maintenance of the project's quality and timeline within its budgetary resources by means of standardized processes, frequent reporting, and performance metrics. In GSD, the challenges are at the level of language incompatibility, time zones, and cross-cultural

misunderstandings that can be overcome by good documentation, clear roles, periodic face-to-face meetings, and instilling a sense of trust and accountability across the distributed teams [34].

E. ACTIVE INVOLVEMENT OF USERS

GSD involves active user participation to overcome the challenges of geographical, cultural, and temporal complications across the distributed teams. That means the software will remain usable, ensuring feedback and clarity of requirements early and throughout the development lifecycle. This involves active participation of the users in elicitation of requirements, iterative reviews, and testing processes. Such collaboration reduces misunderstandings, mitigates the risks of rework, and enhances product quality. It instills trust and aligns the development teams and stakeholders, which is particularly important in GSD, where communication gaps and mismatched expectations can lead to project failure [35].

F. COLLABORATION AMONG TEAM MEMBERS

Collaboration is most crucial in GSD because the teams are distributed across time zones, cultures, and languages. From this respect, effective collaboration on GSD can be achieved through the use of communication tools that establish processes to support and also build cultural awareness for such differences. Miscommunication, delays, and ways of working stand out as common issues that require agility of teams with frequent virtual meetings and high levels of documentation to assure alignment. Additionally, trust-building and establishing shared goals are essential to enhance coordination and productivity in a globally dispersed environment [10], [32], [51].

G. PROFICIENT SOFTWARE DEVELOPERS

The competent software developers in the context of GSD are those with deep technical expertise, good communication skills, and the ability to collaborate on geographically dispersed teams. Developers will have deep knowledge in relevant programming languages, tools, and frameworks of the project, and they will be familiar with principles and best practices in software engineering. They are the best in handling challenges specific to GSD: time zone differences, cultural diversity, and coordination across remote teams. Their adaptability, problem-solving skills, and experience with collaborative tools make them indispensable in the success of GSD projects [48], [49].

H. APPROPRIATE FORMATION OF TEAM

In the context of GSD, appropriate team formation means assembling a group of professionals with complementary skills, diverse cultural backgrounds, and experience across multiple geographies to achieve project goals effectively. This involves a careful choice in terms of technical expertise, communication skills, time-zone differences, and cultural compatibility that foster collaboration and minimize

misunderstandings. Clear role assignments, clear understanding of responsibility at both the individual and collective levels, and a clear leadership setup are necessary for the smoothing of coordination in a distributed environment. Also, collaboration tools and standardized processes regarding knowledge sharing and conflict resolution will help mitigate the geographical and cultural dispersion challenges in GSD teams [42], [49].

I. ROLES AND RESPONSIBILITIES OF TEAM MEMBERS

In GSD, well-defined responsibilities and roles go a long way in reducing the obstacles provided by distance variables such as geographical, cultural, and time zone differences. Well-defined roles make sure that every single member knows what his job is, what deliverables are expected out of him or her, and also to whom these should be delivered, thereby reducing ambiguity and overlap. For example, project managers are responsible for scheduling and resource allocation, developers focus on the coding and implementation aspects, quality assurance engineers are concerned with testing and defect resolution, whereas business analysts take care that requirements fit the stakeholder needs. Well-defined responsibilities ensure accountability of the team members, smooth communication, and collaboration even across dispersed teams, thus enabling better project coordination and increasing success rates in GSD environments [39], [48], [49].

J. STATE OF THE ART TECHNOLOGICAL INFRASTRUCTURE

The state-of-the-art technological infrastructure for members in GSD emphasizes advanced tools and systems that support seamless collaboration across geographically dispersed teams. This includes cloud-based development environments, robust version control systems like Git, continuous integration/continuous deployment (CI/CD) pipelines, and containerization technologies such as Docker and Kubernetes to streamline deployment. In addition, secure communication tools like Slack or Microsoft Teams, together with project management systems such as Jira, Trello, or Asana, ensure enhanced collaboration in real time and task tracking. High-speed internet, cybersecurity protocols, and scalable computing resources ensure reliable and secure access. This infrastructure allows GSD teams to be productive, innovative, and deliver quality software while mitigating the challenges brought about by time zones and cultural diversity [37], [38].

K. FOSTERING CULTURAL AWARENESS

Promoting cultural awareness in the GSD setting includes identifying and understanding the varied cultural backgrounds, standards, communication and working styles of team members spread across various countries and regions. It stresses the significance of adapting management styles, collaboration, and communication strategies to value these cultural differences, confirming effective coordination, reducing misunderstandings, and enhancing overall

productivity. By nurturing cultural sensitivity and inclusivity, organizations can improve collaboration, innovation, and the quality of software products in a GSD setting [33], [50].

L. EFFECTIVE SHARING OF PROJECT KNOWLEDGE

Effective sharing of project knowledge in GSD involves the seamless exchange of information, insights, and expertise among geographically dispersed teams, often across different time zones and cultures. It requires the use of collaborative tools and communication platforms that facilitate real-time interactions and the exchange of documentation, code, and best practices. Successful knowledge sharing also depends on establishing clear processes, fostering a culture of trust and openness, and ensuring that project stakeholders have the necessary resources and support to capture, store, and disseminate valuable knowledge. This ensures alignment, minimizes misunderstandings, and enhances decision-making and project performance [43], [44], [45], [46].

M. MANAGING TIME ZONE DIFFERENCES

Time zone differences in GSD involve coordinating teams that are spread across diverse geographic locations with different work hours. This calls for careful planning to make sure that communication and collaboration remain effective despite the time gaps. Rotating meeting times, using asynchronous communication tools (such as email and project management software), and setting clear deadlines and deliverables are all necessary. Effective time zone management helps minimize delays, ensures continuous progress, and adjusts the varied working hours of team members, developing a collaborative and productive environment [43].

N. CORE PROJECT MANAGEMENT PROCESSES

Core project management processes for GSD encompass the organized planning, execution, monitoring, and control of project tasks in distributed teams. These comprise the definition of project scope, goals, and deliverables, scheduling and resource planning, and risk, quality, and stakeholder communications management. Coordination across time zones and cultures is key, demanding flexible communication plans, collaborative tools, and elaborate documentation. Continuous tracking and adjustments ensure alignment with the project's goals, ensuring that diverse teams work toward a unified outcome while navigating challenges like resource constraints, technical dependencies, and changing requirements [10], [32].

O. TRACKING PROJECT PROGRESS

Progress tracking of projects in GSD entails following the progress of tasks, milestones, and deliverables in geographically dispersed teams. It demands the utilization of collaborative tools, including project management tools and version control systems, to promote transparent communication and coordination across teams situated in various time zones. Regular updates about the status, sprint reviews, and

performance metrics track progress, although challenges pose with time zones, cultural variances, and communication barriers. Effective tracking ensures that the project stays on schedule, meets quality standards, and aligns with the overall business objectives despite the complexities of a global development environment [10], [32], [33].

P. LANGUAGE DIFFERENCES

Language differences in GSD refer to the challenges that arise when team members, stakeholders, or users speak different languages. These differences can affect communication, collaboration, and understanding, leading to misinterpretations, delays, and errors. In such diverse environments, the use of a common language, often English, is common but may still cause difficulties due to varying proficiency levels. Additionally, cultural nuances and idiomatic expressions can further complicate interactions, impacting requirements gathering, documentation, and testing. Effective management of language barriers is crucial to ensuring successful communication and the smooth execution of projects in GSD teams [51], [52], [53], [54].

IV. ANALYSIS AND RESULTS

ISM is a methodical approach that consists of various steps including variable identification, self-structural interaction matrix (SSIM) development with decision makers, reachability matrix (RM) and final reachability matrix (FRM) computation, level partitioning, conical matrix formulation, digraph, and finally an ISM model. The analysis was performed using SmartISM software [12], as discussed in the following paragraphs.

A. IDENTIFICATION OF VARIABLES

The sixteen CSFs were identified in the context of managing GSD projects. The DMs input and a review of the literature were used to identify these CSFs. Table 3 presents list of CSFs of effective project management in GSD. Each CSF is assigned a specific code.

TABLE 3. List of CSFs along with codes.

Codes	CSFs
CSF1	State of the art technological infrastructure
CSF2	Proficient software developers
CSF3	Adeptness in negotiation
CSF4	Effective coordination & control
CSF5	Productive communication skills
CSF6	Frequent communication
CSF7	Active involvement of users
CSF8	Collaboration among team members
CSF9	Appropriate formation of team
CSF10	Roles and responsibilities of team members
CSF11	Tracking project progress
CSF12	Core project management processes
CSF13	Effective sharing of project knowledge
CSF14	Managing time zone differences
CSF15	Language differences
CSF16	Fostering cultural awareness

B. STRUCTURAL SELF-INTERACTION MATRIX (SSIM)

This matrix represents the relationships that the DMs discovered during the group decision-making process. The variables have been compared and several possible associations are assigned to them which are represented by the letters V, A, X, and O. The letter V denotes that the first influencing the second, A indicates that the second influencing the first, X signifies mutual influence, and O indicates no link. Because the subsequent mathematical processes rely on this matrix, this ISM phase is the most important. For example, CSF9 (appropriate formation of team) influences CSF 11 (tracking project progress) and similarly by CSF14, CSF15, and CSF16, are represented by 'V'. It is influenced by CFS12 (core project management processes) and similarly by CSF13, represented by 'A'. Moreover, CSF9 has a mutual influence with CSF10 (roles and responsibilities of team members), represented by 'X' as shown in row 9 of Table 4. The correlations presented in Table 4 were determined by closely examining each of the 120 pairs of CSFs.

C. REACHABILITY MATRIX (RM) AND FINAL REACHABILITY MATRIX (FRM)

The RM is constructed based on the principle that if a variable (listed in the SSIM rows) influences or leads to another variable (listed in the SSIM columns), it is represented by one. All of the letters in the SSIM, including V, A, X, and O, are transformed to 0 or 1 using the following logic, where i and j represent the row and column indices, respectively. As a result, the RM is created.

If $SSIM(i, j) = V$, then $RM(i, j) = 1$ and $RM(j, i) = 0$;
 If $SSIM(i, j) = A$, then $RM(i, j) = 0$ and $RM(j, i) = 1$;
 If $SSIM(i, j) = X$, then $RM(i, j) = 1$ and $RM(j, i) = 1$;
 If $SSIM(i, j) = O$, then $RM(i, j) = 0$ and $RM(j, i) = 0$;

Afterward, transitive reachability relations are taken into consideration while constructing the FRM. For instance, if variable a affects variable b and variable b affects variable c , then variable a also affects variable c . This is an example of a transitive relation. Additionally, higher-order transitive relations are observed, such as those in which a effects b through two or more variables. These relationships are denoted by 1^* in the FRM (Table 5). The number of non-zero cells in the rows and columns for each factor represents the driving and dependence powers of each variable, which are also shown in Table 5.

D. LEVEL PARTITIONING (LP)

The level partitioning is a sophisticated process, it arranges variables in a vertical form. The function iterates until each variable's level is determined. For every variable, its antecedent set (variables that influence it) and reachability set (variables it can influence) are determined. Next, the intersection of these two sets is calculated. Variables with identical intersection and reachability sets are assigned the highest level and removed from further iterations. After nine

TABLE 4. Structural self-interaction matrix (SSIM).

Critical Success Factors	1	6	5	4	3	2	1	1	0	9	8	7	6	5	4	3	2	1
CSF1	O	V	V	V	V	V	V	V	V	V	V	V	V	V	V	O	O	
CSF2	V	V	V	O	O	V	V	V	V	V	V	V	V	V	V	V		
CSF3	V	V	V	O	O	V	V	V	V	V	V	V	V	V	V			
CSF4	V	V	V	A	A	V	A	A	X	A	A	A						
CSF5	V	V	V	A	O	V	V	V	V	V	V							
CSF6	V	V	V	A	A	V	V	V	V	X								
CSF7	V	V	V	A	A	V	V	V	V									
CSF8	V	V	V	A	A	V	A	A										
CSF9	V	V	V	A	A	V	X											
CSF10	V	V	V	A	A	V												
CSF11	A	A	A	A	A													
CSF12	V	V	V	V														
CSF13	V	V	V															
CSF14	O	O																
CSF15	X																	
CSF16																		

iterations, the 16 CSFs were categorized into nine levels. The allocated levels for each CSFs are shown in Table 6.

E. CONICAL MATRIX, DIGRAPH, AND FINAL ISM MODEL

The conical matrix is a kind of FRM that is arranged both horizontally and vertically based on the levels of variables. This matrix is used to develop the digraph. The digraph is a directed graph where the edges are ordered based on their relationships and the nodes are represented by variable IDs. To improve the clarity of the final ISM model, unnecessary transitive relationships are removed from the digraph. To make sure that the ISM model is completely valid, an additional analysis was conducted through DMs. If the derived ISM model does not meet the DM's expectations, the SSIM is re-examined, and all procedures are re-evaluated to regenerate the model. In this study, two adjustments were made to the overall analysis, the final ISM model is shown in Figure 1.

F. MICMAC ANALYSIS

The MICMAC analysis separates the variables into four groups based on the driving and reliance powers of the variables: autonomous, dependent, linkage, and independent variables. This division is based on the multiplication features of matrices. Autonomous and linkage variables do not exist in this model (Figure 2). All CSFs have been categorized either as dependent or independent and are located in quadrants II or IV respectively. The most independent variable identified is state of the art technological infrastructure, while the most dependent variable is tracking project progress.

TABLE 5. Final reachability matrix (FRM).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Driving Power
CSF 1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	14
CSF 2	0	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	13
CSF 3	0	0	1	1	1	1	1	1	1	1	0	0	1	1	1	1	12
CSF 4	0	0	0	1	0	0	0	1	0	0	1	0	0	1	1	1	6
CSF 5	0	0	0	1	1	1	1	1	1	1	0	0	1	1	1	1	11
CSF 6	0	0	0	1	0	1	1	1	1	1	1	0	0	1	1	1	10
CSF 7	0	0	0	1	0	1	1	1	1	1	1	0	0	1	1	1	10
CSF 8	0	0	0	1	0	0	0	1	0	0	1	0	0	1	1	1	6
CSF 9	0	0	0	1	0	0	0	1	1	1	1	0	0	1	1	1	8
CSF 10	0	0	0	1	0	0	0	1	1	1	1	0	0	1	1	1	8
CSF 11	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1
CSF 12	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	13
CSF 13	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	1	12
CSF 14	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	2
CSF 15	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	3
CSF 16	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	3
Dependence Power	1	1	2	1	6	8	8	1	1	1	1	2	3	1	1	1	
				2			2	0	0	6			3	4	4		

V. MODEL VALIDATION

For the validation of the proposed model, six professionals having varying experiences in the management of GSD projects were approached. These professionals include senior managers (02) project managers (02) team lead (01) and academician (01) having total IT industry experience of 20 years, 19 years, 15 years, 13 years, 10 years and 20 years respectively. The model was presented and elucidated to them with the definitions of variables as desired. The duration of in-depth interviews was in the range of twenty minutes to one hour time. The in-depth interviews were also followed by multiple feedback rounds for further validation. The transcription of summarized, anonymous, and randomized interviews is described below.

An interview with one of the reviewers was remain continued for an hour and he shared very important insights regarding the successful management of GSD projects. He suggested that the model should clearly represent four layers, starting from the infrastructure, communication, collaboration and eventually project progress. He also stressed on the triangular interaction loop between developer, knowledge, management and project progress. He disagreed

TABLE 6. Level partitioning.

Elements (Mi)	Reachability Set R(Mi)	Antecedent Set A(Ni)	Intersection Set R(Mi)∩A(Ni)	Level
1	1	1	1	9
2	2	2	2	8
3	3	2, 3	3	7
4	4, 8	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13	4, 8	3
5	5	1, 2, 3, 5, 12, 13	5	6
6	6, 7	1, 2, 3, 5, 6, 7, 12, 13	6, 7	5
7	6, 7	1, 2, 3, 5, 6, 7, 12, 13	6, 7	5
8	4, 8	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13	4, 8	3
9	9, 10	1, 2, 3, 5, 6, 7, 9, 10, 12, 13	9, 10	4
10	9, 10	1, 2, 3, 5, 6, 7, 9, 10, 12, 13	9, 10	4
11	11	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16	11	1
12	12	1, 12	12	8
13	13	1, 12, 13	13	7
14	14	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14,	14	2
15	15, 16	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 16,	15, 16	2
16	15, 16	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 16	15, 16	2

with the fostering cultural awareness; his view is that if team is properly formed with right members, then cultural awareness has little importance. He may have only observed four layers of the proposed model but did not have time to discuss and clarify each variable within the model. Furthermore, he demonstrated mainly the independent variables of the model.

The second reviewer provided valuable feedback, highlighting that properly managing time zone differences, language differences and cultural awareness fosters productive communication, coordination and collaboration. He has given importance to cultural awareness because it helps develop working relationship among team members and communicate frequently, freely and informally regarding a project. He has suggested to merge the effective coordination & control and collaboration among team members and rename as coordination, control and collaboration. Another reviewer was fully agreed with the model, except that actively involving user is not an easy endeavor, it requires a lot of work beforehand. The next reviewer also found the model completely appropriate to successfully manage the GSD projects. Moreover, he suggested that coordination and control and collaboration among team members can independently produce varied positive effects over the efficient management. For example, effective coordination and control helps to achieve project objectives and make it certain that project is delivering good quality

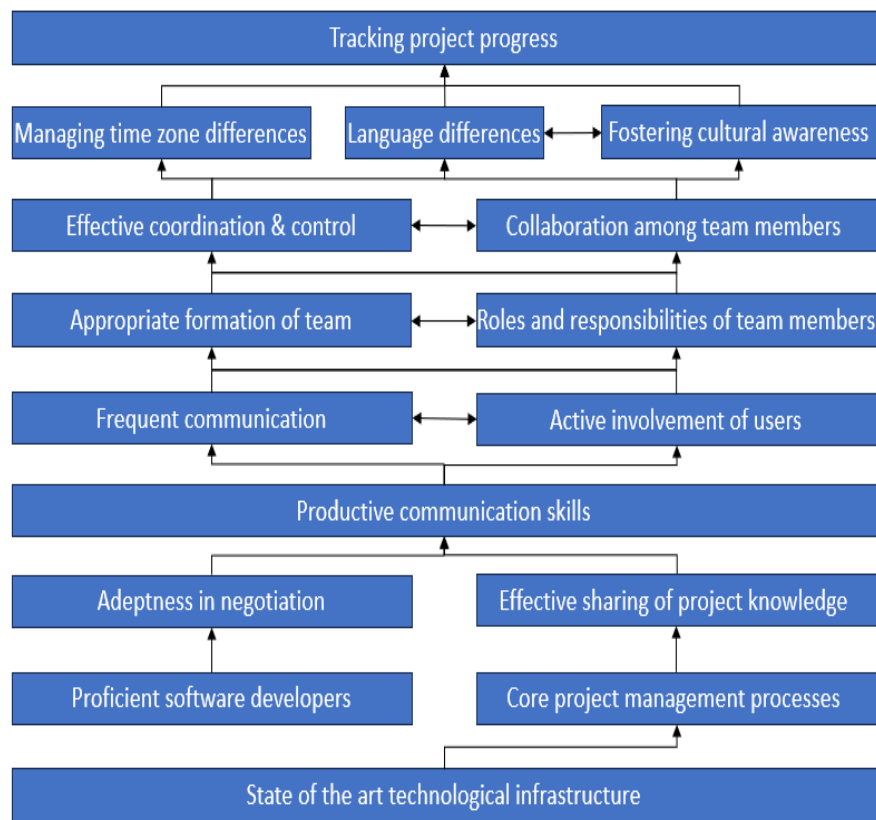


FIGURE 1. ISM model.

results whereas collaboration encourage team members to generate, share ideas, thoughts and solutions through brainstorming. He proposed that adeptness in negotiation should have an element of conflict management so it can be change to adeptness in negotiation and conflict management. The fifth reviewer showed no observation regarding any element of the model. He showed complete agreement with it.

After summarizing reviewers' views, it can be concluded that the model thoroughly characterizes the main parameters of successful management of GSD projects. Generally, reviewers are in a concurrence with the inter-relationships of the model variables. No major concern raised up by the reviewers which necessitates tweaking in the model.

VI. DISCUSSION

To streamline the interrelationship of sixteen CSFs of successful management of GSD projects, the model is elucidated in the following layers: Each layer emphasizes on a particular aspect of GSD that ensures continuous collaboration, efficiency and quality throughout the life of a project. These layers are technological infrastructure, human resource expertise, communication, collaboration and coordination, management and documentation. Technological infrastructure refers to the technological tools and platforms that devise the backbone to perform GSD projects. As regard to GSD, rich technological infrastructure confirms seamless collaboration and access to resources across geographically

distributed teams. The robust infrastructure enables effective communication and collaboration which are key to the successful execution of GSD projects, especially when the team spans at diverse sites. Frequent and structured communication aids in dealing with conflicts, issues and keeping the project aligned to the vision, and exchanging updates. The next layer emphasizes on the proficiency of software developers and other stakeholders involved in the GSD project. The success of GSD project rests on the collective proficiency and ability to work together seamlessly despite geographical limitations. Managing cultural, language and time zone differences are especially important in GSD projects, as project teams often span across various countries and regions and face cultural, time zone and language barriers. By incorporating cultural sensitivity and effective management of language and time zone differences, this layer ensures a balanced, coordinated and constructive team environment, which is central to successfully manage a GSD project. The documentation and knowledge sharing are important to the smooth operation and success of a GSD project. They confirm consistency, good decision making, mitigate risks, improve quality and develop a culture of collaboration, eventually leading to more efficient and effective delivery of a GSD project.

VII. CONCLUSION, LIMITATIONS, AND FUTURE WORK

In the current era of digitalization, software development work is being extensively performed beyond the bound-

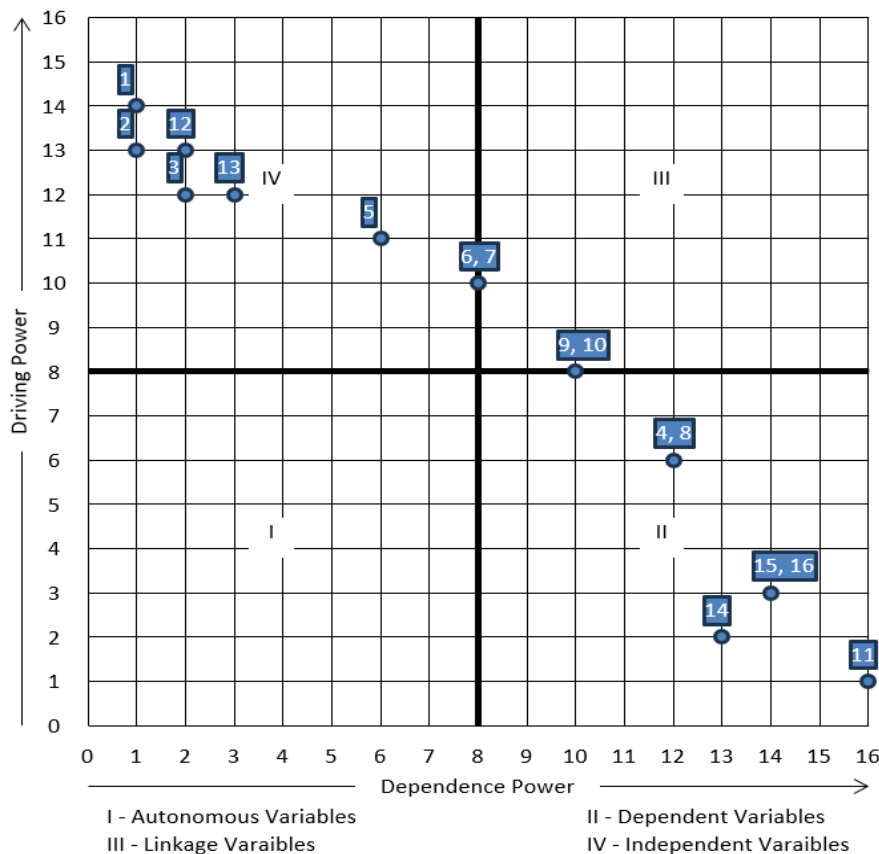


FIGURE 2. MICMAC diagram.

aries. However, it is imperative to ensure the successful management of GSD projects. This research study offers computation, prioritization and classification of CSFs of software project management in GSD using ISM and MICMAC. The ISM model proposed in this study can assist the management of GSD projects in comprehending the inter-relationships among CSFs for the formulation of management strategy, decision making, project planning and optimization. The management of GSD setting can achieve the successful completion of a project by carefully administering the independent CSFs including technological infrastructure, proficient developers, negotiation, communication, coordination and control, collaboration, user involvement, right team, defined role and responsibilities of team members, tracking project progress, core management processes and knowledge sharing.

The strong presence of such independent CSFs may assist in improving the dependent CSFs such as time zone differences, language differences and cultural awareness, which are fundamental to the successful management of GSD projects. This may eventually lead to the dynamics of project management in GSD setting with regard to improved interaction with project manager and/or team lead and team members of a GSD project.

ISM uses a group decision-making approach to identify contextual relationships between variables, thus prone to individual biases. However, validation was performed by involving six expert reviewers. Their assessment verified that the model indeed represents significant dimensions of GSD project management and their interrelations well.

The proposed ISM based model improves the existing work by not only identifying and ranking CSFs but also analyzing their interrelationships in a systematic manner. Moreover, this study reveals the hierarchical nature and driving dependency relationships among factors, assisting practitioners to concentrate on key enablers of project success.

This research has certain limitations such as it involved limited number of expert professionals, which may affect the generalizability of the results. The generalization of the findings could be improved with more participants. Secondly, the proposed study has relied on a single validation method through expert validation. Additional validation methods including case studies, empirical testing through statistical analysis or industry surveys could further improve the reliability and generalizability of the findings of this research. The future studies might investigate alternative strategies to extend the comprehensiveness of the findings.

ACKNOWLEDGMENT

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP2/179/45.

REFERENCES

- [1] P. Björndal, K. Smiley, and P. Mohapatra, "Global software project management: A case study," in *Proc. Int. Conf. Softw. Eng. Approaches Offshore Outsourced Develop.*, vol. 4, St. Petersburg, Russia. Berlin, Germany: Springer, Jun. 2010, pp. 64–70.
- [2] J. Noll, S. Beecham, and I. Richardson, "Global software development and collaboration: Barriers and solutions," *ACM Inroads*, vol. 1, no. 3, pp. 66–78, Sep. 2010.
- [3] A. Manjavacas, A. Vizcaíno, F. Ruiz, and M. Piattini, "Global software development governance: Challenges and solutions," *J. Softw., Evol. Process*, vol. 32, no. 10, p. e2266, Oct. 2020.
- [4] N. Saleem, "Empirical analysis of critical success factors for project management in global software development," in *Proc. ACM/IEEE 14th Int. Conf. Global Softw. Eng. (ICGSE)*, May 2019, pp. 68–71.
- [5] M. Niazi, S. Mahmood, M. Alshayeb, A. M. Qureshi, K. Faisal, and N. Cerpa, "Toward successful project management in global software development," *Int. J. Project Manage.*, vol. 34, no. 8, pp. 1553–1567, Nov. 2016.
- [6] H. N. N. Mohd and S. Shamsul, "Critical success factors for software projects: A comparative study," *Sci. Res. Essays*, vol. 6, no. 10, pp. 2174–2186, May 2011.
- [7] V. Garousi, A. Tarhan, D. Pfahl, A. Coskunçay, and O. Demirörs, "Correlation of critical success factors with success of software projects: An empirical investigation," *Softw. Qual. J.*, vol. 27, pp. 429–493, 2019.
- [8] S. Siddique, M. Naveed, A. Ali, I. Keshta, M. I. Satti, A. Irshad, Z. Alomari, O. C. Edo, and O. A. Diekola, "An effective framework to improve the managerial activities in global software development," *Nonlinear Eng.*, vol. 12, no. 1, Sep. 2023, Art. no. 20220312.
- [9] M. A. Akbar, A. A. Khan, S. Mahmood, and K. Smolander, "Successful management of cloud-based global software development projects: A multivocal study," *J. Softw., Evol. Process*, vol. 36, no. 4, p. e2527, Apr. 2024.
- [10] M. El Bajta, A. Idri, J. N. Ros, J. L. Fernández-Alemán, J. M. Carrillo de Gea, F. García, and A. Toval, "Software project management approaches for global software development: A systematic mapping study," *Tsinghua Sci. Technol.*, vol. 23, no. 6, pp. 690–714, Dec. 2018.
- [11] J. Alqahtani, A. Siddique, A. M. Aseere, A. Alasiry, and Q. N. Naveed, "Evaluating success factors of software project management in global software development," *IEEE Access*, vol. 12, pp. 22345–22358, 2024.
- [12] N. Ahmad and A. Qahmash, "SmartISM: Implementation and assessment of interpretive structural modeling," *Sustainability*, vol. 13, no. 16, p. 8801, Aug. 2021.
- [13] N. Ahmad, N. Hoda, and F. Alahmari, "Developing a cloud-based mobile learning adoption model to promote sustainable education," *Sustainability*, vol. 12, no. 8, p. 3126, Apr. 2020.
- [14] Q. N. Naveed, N. Ahmad, S. Qamar, N. Khan, A. Naim, M. R. Hussain, M. R. N. Qureshi, A. O. Alsayed, and K. Mohiuddin, "Relationship modeling for OSN-based E-learning deployment," in *Proc. IEEE 6th Int. Conf. Eng. Technol. Appl. Sci. (ICETAS)*, Dec. 2019, pp. 1–7.
- [15] Q. N. Naveed, M. R. N. Qureshi, A. O. Alsayed, N. Ahmad, S. Sanobar, and A. Shah, "Assimilating E-learning barriers using an interpretive structural modeling (ISM)," in *Proc. 4th IEEE Int. Conf. Eng. Technol. Appl. Sci. (ICETAS)*, Nov. 2017, pp. 1–7.
- [16] N. Ahmad, "The structural modeling of significant factors for sustainable cloud migration," *Int. J. Intell. Eng. Syst.*, vol. 14, no. 2, pp. 1–7, 2021.
- [17] J. Dobbins and R. Donnelly, "Critical success factor analysis for technology acquisition program management," in *Proc. PMI Res. Conf.*, London, U.K., Jul. 2004.
- [18] B. B. Chua and T. S. Ong, "Critical success factors for global software development projects: A systematic review," *J. Softw. Eng. Appl.*, vol. 8, no. 5, pp. 269–283, 2015.
- [19] J. G. Guzmán, J. S. Ramos, A. A. Seco, and A. S. Esteban, "Success factors for the management of global virtual teams for software development," *Int. J. Human Capital Inf. Technol. Professionals*, vol. 2, no. 2, pp. 48–59, Apr. 2011.
- [20] H. Cicibas, O. Unal, and K. A. Demir, "A comparison of project management software tools (PMST)," in *Proc. Int. Conf. Softw. Eng. Res. Pract.* Las Vegas, NV, USA: SERP, Jul. 2010, pp. 560–565.
- [21] M. Niazi, S. Mahmood, M. Alshayeb, M. R. Riaz, K. Faisal, and N. Cerpa, "Challenges of project management in global software development: Initial results," in *Proc. Sci. Inf. Conf.*, Oct. 2013, pp. 202–206.
- [22] M. Korkala, M. Pikkariainen, and K. Conboy, "A case study of customer communication in globally distributed software product development," in *Proc. 11th Int. Conf. Product Focused Softw.*, Jun. 2010, pp. 43–46.
- [23] R. Jain and U. Suman, "A project management framework for global software development," *ACM SIGSOFT Softw. Eng. Notes*, vol. 43, no. 1, pp. 1–10, Mar. 2018.
- [24] N. Saleem, S. Mathrani, and N. Taskin, "Investigating critical success factors of project management in global software development: A work in progress," *Decis. Support Syst.*, vol. 14, no. 2, pp. 1–15, 2019, doi: 10.1016/j.dss.2008.10.005. [Online]. Available: https://www.researchgate.net/profile/Alireza_Amrollahi/publication/264314726_How_Open_Source_Softwa
- [25] M. Radujković and M. Sjekavica, "Project management success factors," *Proc. Eng.*, vol. 196, pp. 607–615, Jan. 2017.
- [26] I. Nurdiani, R. Jabangwe, D. Šmite, and D. Damian, "Risk identification and risk mitigation instruments for global software development: Systematic review and survey results," in *Proc. IEEE 6th Int. Conf. Global Softw. Eng. Workshop*, Aug. 2011, pp. 36–41.
- [27] S. Y. Chadli, A. Idri, J. L. Fernández-Alemán, J. N. Ros, and A. Toval, "Identifying risks of software project management in global software development: An integrative framework," in *Proc. IEEE/ACS 13th Int. Conf. Comput. Syst. Appl. (AICCSA)*, Nov. 2016, pp. 1–7.
- [28] M. Marinho, J. Noll, and S. Beecham, "Uncertainty management for global software development teams," in *Proc. 11th Int. Conf. Quality Inf. Commun. Technol. (QUATIC)*, Sep. 2018, pp. 238–246.
- [29] J. Lim, "Examining an effective project management office model for global software development environments," Ph.D. dissertation, College Manag. Technol., Walden Univ., Minneapolis, MN, USA, 2012.
- [30] S. Misra, R. Colomo-Palacios, P. Soto-Acosta, and T. Pusatli, "A discussion on the role of people in global software development," *Tech. Gazette*, vol. 20, no. 3, pp. 525–531, 2013.
- [31] S. Y. Chadli, A. Idri, J. N. Ros, J. L. Fernández-Alemán, J. M. C. de Gea, and A. Toval, "Software project management tools in global software development: A systematic mapping study," *SpringerPlus*, vol. 5, no. 1, pp. 1–38, Dec. 2016.
- [32] M. Yaseen, M. Ali, A. Ur, S. Nabi, S. Khan, and M. Bacha, "Practices for effective software project management in global software development: A systematic literature review," *Int. J. Comput. Appl.*, vol. 177, no. 36, pp. 1–6, Feb. 2020.
- [33] U. Remus and M. Wiener, "Critical success factors for managing offshore software development projects," *J. Global Inf. Technol. Manage.*, vol. 12, no. 1, pp. 6–29, Jan. 2009.
- [34] S. Asiri and M. R. Jameel Qureshi, "The proposed GSD model to solve coordination and communication problems," *Int. J. Mod. Educ. Comput. Sci.*, vol. 6, no. 11, pp. 25–30, Nov. 2014.
- [35] V. Bhoola, "Impact of project success factors in managing software projects in india: An empirical analysis," *Bus. Perspect. Res.*, vol. 3, no. 2, pp. 109–125, Jul. 2015.
- [36] P. M. Bosch-Sijtsema and M. Vartiainen, "Getting it done: Critical success factors for project managers in dispersed work settings," *Int. J. Project Manage.*, vol. 31, no. 1, pp. 67–79, 2013.
- [37] S. Laato, M. Mäntymäki, A. K. M. N. Islam, S. Hyrynsalmi, and T. Birkstedt, "Trends and trajectories in the software industry: Implications for the future of work," *Inf. Syst. Frontiers*, vol. 25, pp. 929–944, Apr. 2022.
- [38] A. M. Pereira, R. Q. Goncalves, C. G. Von Wangenheim, and L. Buglione, "Comparison of open-source tools for project management," *Int. J. Softw. Eng. Knowl. Eng.*, vol. 23, no. 2, pp. 189–209, 2013.
- [39] A. Ahmad, "Effective distribution of roles and responsibilities in global software development teams," M.S. thesis, School Comput., Blekinge Inst. Technol., Karlskrona, Sweden, 2012.
- [40] V. Casey and I. Richardson, "Practical experience of virtual team software development," in *Proc. Eur. Soft. Process Improvement (Euro SPI)*, Trondheim, Norway, 2004, p. 3-D.9.

- [41] A. E. Akgün, "Team wisdom in software development projects and its impact on project performance," *Int. J. Inf. Manage.*, vol. 50, pp. 228–243, Feb. 2020.
- [42] A. Costa, F. Ramos, M. Perkusich, E. Dantas, E. Dilenzo, F. Chagas, A. Meireles, D. Albuquerque, L. Silva, H. Almeida, and A. Perkusich, "Team formation in software engineering: A systematic mapping study," *IEEE Access*, vol. 8, pp. 145687–145712, 2020.
- [43] M. Noor and Z. A. Rana, "Towards better knowledge management in global software engineering," in *Proc. 4th Int. Conf. Comput. Inf. Sci. (ICCOINS)*, Aug. 2018, pp. 1–6.
- [44] S. G. Shiva, S. B. Lee, L. A. Shala, and C. B. Simmons, "Knowledge management in global software development," *Int. J. Distrib. Sensor Netw.*, vol. 5, no. 1, p. 6, 2009.
- [45] M. Zahedi, M. Shahin, and M. Ali Babar, "A systematic review of knowledge sharing challenges and practices in global software development," *Int. J. Inf. Manage.*, vol. 36, no. 6, pp. 995–1019, Dec. 2016.
- [46] E. H. M. Huzita, G. C. L. Leal, R. Balancieri, T. F. C. Tait, E. Cardoza, R. R. d. M. Penteado, and R. L. Vivian, "Knowledge and contextual information management in global software development: Challenges and perspectives," in *Proc. IEEE 7th Int. Conf. Global Softw. Eng. Workshops*, Aug. 2012, pp. 43–48.
- [47] A. Taweel and P. Brereton, "Modelling software development across time zones," *Inf. Softw. Technol.*, vol. 48, no. 1, pp. 1–11, Jan. 2006.
- [48] A. Siddique, G. M. Butt, A. Zahid, Q. N. Naveed, and M. T.-H. Alouane, "Analyzing software industry trends to improve curriculum," *IEEE Access*, vol. 12, pp. 22510–22523, 2024.
- [49] A. Siddique, Q. Noorulhasan Naveed, N. Kraiem, and M. Aref Abdul Rasheed, "Investigating critical success factors for effective management of IT projects," *IEEE Access*, vol. 12, pp. 166717–166729, 2024.
- [50] J. Marinho, M. Luna, and S. Beecham, "Global software development: Practices for cultural differences," in *Proc. 19th Int. Conf. Product-Focused Softw. Process Improvement (PROFES)*. Wolfsburg, Germany: Springer, vol. 2018, pp. 299–317.
- [51] A. Niva, J. Markkula, and E. Annanperä, "Junior software engineers' international communication and collaboration competences," *IEEE Access*, vol. 11, pp. 139039–139068, 2023.
- [52] Y. Y. Jusoh, R. N. Haizan Nor, B. A. Mahmood, M. T. Wafeeq, M. A. Ali, and M. N. Baihaqi Jusoh, "Communication management in global software development projects," in *Proc. 4th Int. Conf. Inf. Retr. Knowl. Manage. (CAMP)*, Kota Kinabalu, Malaysia, Mar. 2018, pp. 1–7.
- [53] B. Lutz, "Linguistic challenges in global software development: Lessons learned in an international SW development division," in *Proc. 4th IEEE Int. Conf. Global Softw. Eng.*, Limerick, Ireland, Jul. 2009, pp. 249–253.
- [54] Y. Wang, Y. Yue, W. Wang, and G. Zhang, "Uncovering non-native speakers' experiences in global software development teams—A bourdieusian perspective," in *Computer Supported Cooperative Work (CSCW)*, vol. 1, 2024, pp. 1–36.



ANSAR SIDDIQUE received the Ph.D. degree from International Islamic University Islamabad, Pakistan. Currently, he is a Senior Associate Professor and a Postgraduate Programs Coordinator with the Department of Computer Sciences, Bahria University, Lahore Campus (BULC), Pakistan. Additionally, he leads a research group focused on software engineering and human–computer interaction. He has extensive experience in teaching, research, and educational administration. He has published his research in various peer-reviewed journals and international conference proceedings. He also serves as a reviewer for several reputable international journals and IEEE sponsored conferences. His research interests include e-learning, usability and HCI, software engineering, and the application of artificial intelligence (AI) in education and software engineering.



NOORULHASAN NAVEED QUADRI received the Ph.D. degree in information technology from the Department of Information Systems, Kuliyah of Information and Communication Technology (KICT), International Islamic University Malaysia (IIUM), Kuala Lumpur. He was an IT Engineer with Saudi Aramco and Bank, Riyadh, Saudi Arabia. He is currently teaching with the College of Computer Science, King Khalid University, Saudi Arabia. He has many publications in refereed/indexed international journals and the IEEE, ACM, and Scopus Springer-sponsored conferences. His current research interests include e-learning, m-learning, cloud computing, cloud-based e-learning systems, and technology-enhanced learning. He is a reviewer and an international advisory board of several conferences and journals.



NAIM AHMAD received the M.S. degree in computer and information sciences from the University of South Alabama, Mobile, AL, USA, and the Ph.D. degree in enterprise systems from Aligarh Muslim University, India. He worked in USA as a Research Assistant with U.S. DOE-funded project and as a Graduate Assistant. He has taught as an Assistant Professor with the Management and Engineering Colleges, India, for seven years. He has been a Faculty Member with King Khalid University, Saudi Arabia, since 2013. He has publications in refereed international journals, conferences, and book chapters. His current research interests include enterprise systems, e-learning, sustainability, data science, and MCDM. He received the Upsilon Pi Epsilon Award for Computing Excellence, USA, in 2003. He has developed Smart ISM Software, globally used for interpretive structural modeling.



HASHIM ELSHAFIE (Member, IEEE) received the bachelor's degree in electronics engineering with a focus on telecommunications from Sudan University of Science and Technology (SUST), and the master's and Ph.D. degrees in electrical engineering with a specialization in telecommunications from Universiti Teknologi Malaysia (UTM). From 2013 to 2015, he was a Senior Researcher with the MIMOS Center of Excellence in Telecommunications Engineering MIMOS-UTM. In addition, he was a member of the Telematic Research Group TRG, UTM. He is an Expert in telecommunications engineering. Moreover, he gained industrial experience with Malaysian Institute of Micro-electronic Systems (MIMOS Berhad), a National Applied Research and Development Centre, Malaysia, from 2011 to 2013. Since January 2016, he has been an Assistant Professor with the Department of Computer Engineering, King Khalid University (KKU), Saudi Arabia. His research interests include telecommunication and networking engineering, satellite and radar communications, antenna engineering and electromagnetic waves, wireless sensor networks, the Internet of Things (IoT), 5G/6G cellular systems, artificial intelligence (AI), spectrum sharing with cognitive radio techniques, and TV white space management schemes.

...